

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

Real SGCC Data Validation Report

1. Dataset Identification

- **File:** D:\wattmaticdataset\data.csv
- **Dimensions:** 42372 rows, 1036 columns
- **CONS_NO Uniqueness:** 42372 unique IDs (Duplicates: 0)
- **FLAG Distribution:** {0: 38757, 1: 3615}
- **Number of Consumption Columns:** 1034
- **Date Parsing Success:** True
- **Earliest Date:** 2014-01-01
- **Latest Date:** 2016-10-31

2. Chronological Order

- **Is Chronologically Sorted?:** False
- **Is Lexicographically Sorted?:** True
- **Current First 20 Columns:**

['2014/1/1', '2014/1/10', '2014/1/11', '2014/1/12', '2014/1/13', '2014/1/14', '2014/1/15', '2014/1/16', '2014/1/17', '2014/1/18', '2014/1/19', '2014/1/2', '2014/1/20', '2014/1/21', '2014/1/22', '2014/1/23', '2014/1/24', '2014/1/25', '2014/1/26', '2014/1/27']

- **Chronological First 20 Columns:**

['2014/1/1', '2014/1/2', '2014/1/3', '2014/1/4', '2014/1/5', '2014/1/6', '2014/1/7', '2014/1/8', '2014/1/9', '2014/1/10', '2014/1/11', '2014/1/12', '2014/1/13', '2014/1/14', '2014/1/15', '2014/1/16', '2014/1/17', '2014/1/18', '2014/1/19', '2014/1/20']

Note: The dataset columns are ordered lexicographically (alphabetically by string) rather than chronologically by actual date (e.g., Jan 1st of different years appear consecutively).

3. Data Quality

- **Total Missing Cells:** 11,233,528 (25.64%)
- **FLAG=0 Missingness:** 25.10%

- FLAG=1 Missingness: 31.47%
- **Total Zeros:** 5,788,603 (13.21%)
- FLAG=0 Zeros: 13.67%
- FLAG=1 Zeros: 8.31%
- **Negative Values:** 0
- **Infinite Values:** 0
- **Duplicate Rows:** 0

4. Raw Distribution (Non-NaN values)

Statistic	Overall	FLAG=0 (Normal)	FLAG=1 (Theft)
Min	0.00	0.00	0.00
Max	800003.32	800003.32	514991.78
Mean	9.26	7.70	27.53
Median	4.59	4.43	7.01
Std Dev	273.22	245.39	493.37
5th Pct	0.00	0.00	0.00
95th Pct	25.24	23.05	73.73

5. Target Leakage Audit

- Unlike the synthetic dataset, the minimum consumption values for Normal (0.00) and Theft (0.00) are statistically identical.
- Zeros (13.67% vs 8.31%) and Missing values (25.10% vs 31.47%) exhibit slightly different rates between classes, representing behavioral signals rather than deterministic data-generator artifacts.
- No obvious deterministic proxy for FLAG is present in the raw data matrix.

6. Comparison Against Synthetic Data

- **Synthetic Data:** 1,500 rows. Zeroes and missing values were hardcoded deterministic patterns. Perfect separation was trivially achieved.
- **Real SGCC Data:** 42,372 rows. Contains genuine noise, non-chronological columns, and real-world missingness (e.g., communication failures rather than synthetic drops).

7. Feature Pipeline Compatibility (Required Changes)

The existing 55-feature pipeline (run_fe_extractor.py) must be modified before processing this data:

1. **Chronological Sorting:** The pipeline currently relies on np.diff(X) for volatility features (daily changes) and polyfit for linear trend. Since columns in this real dataset are ordered lexicographically (e.g., 2014/1/1, 2014/1/10), applying diff() right now will compute nonsense transitions between arbitrary dates. The dataset columns must be sorted chronologically first.
2. **Missing Value Handling:** The current pipeline uses ffill(axis=1).bfill(axis=1). However, if columns are not chronological, ffill will propagate data across non-adjacent temporal blocks.
3. **Monthly Aggregations:** Handled safely because dates are parsed independently by pandas.